Are U.S. Import Tariff Burdens Different Across Commodity Categories?

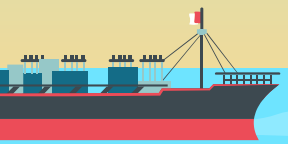
By: ShiHan



Data Information

Source: U.S. Census Bureau monthly import statistics

time	YYYY-MM of the observation
I_COMMODITY	HS-2 commodity chapter code
I_COMMODITY_SDESC	Commodity chapter description
GEN_VAL_MO	General import value
CAL_DUT_MO	Calculated duties collected during the month
CON_VAL_MO	Consumption import customs value during the month
DUT_VAL_MO	Dutiable import value





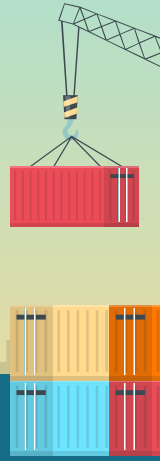
Feature Engineering

$$ETR_{con} = \frac{CAL_DUT_MO}{CON_VAL_MO}$$

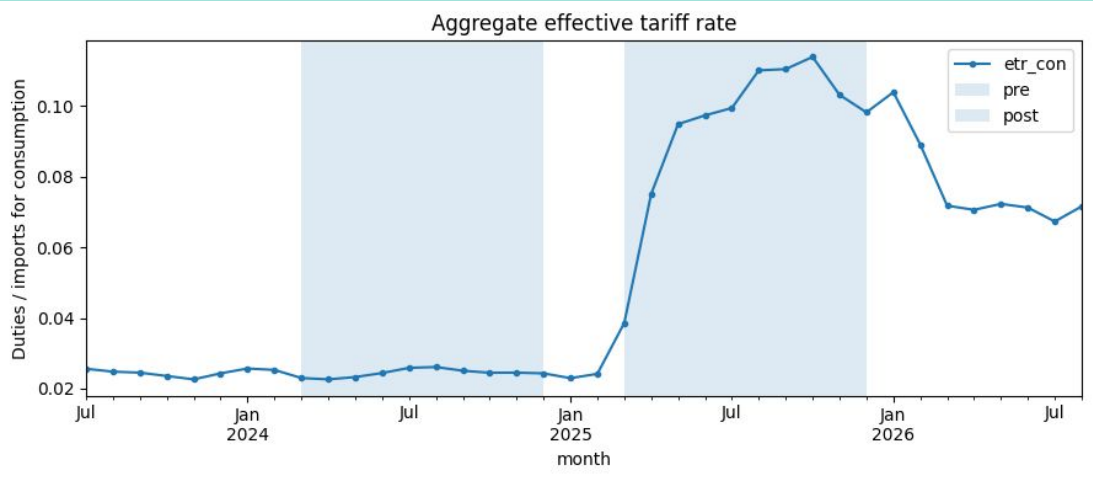
Before Period:
March 2024 to December 2024

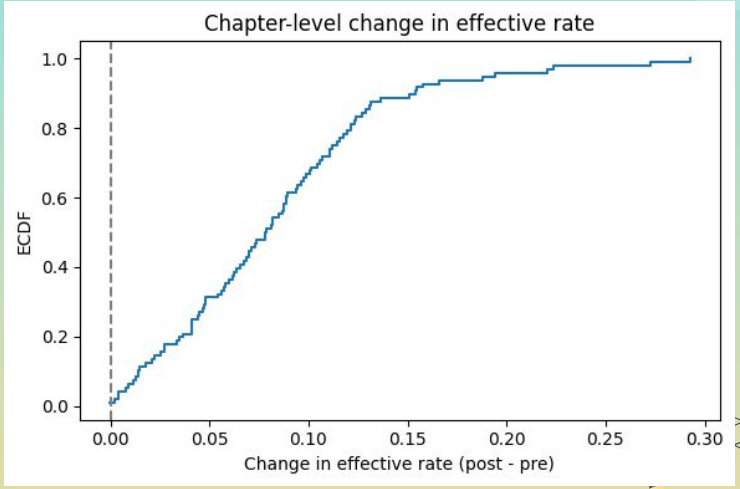
After Period:
March 2025 to December 2025



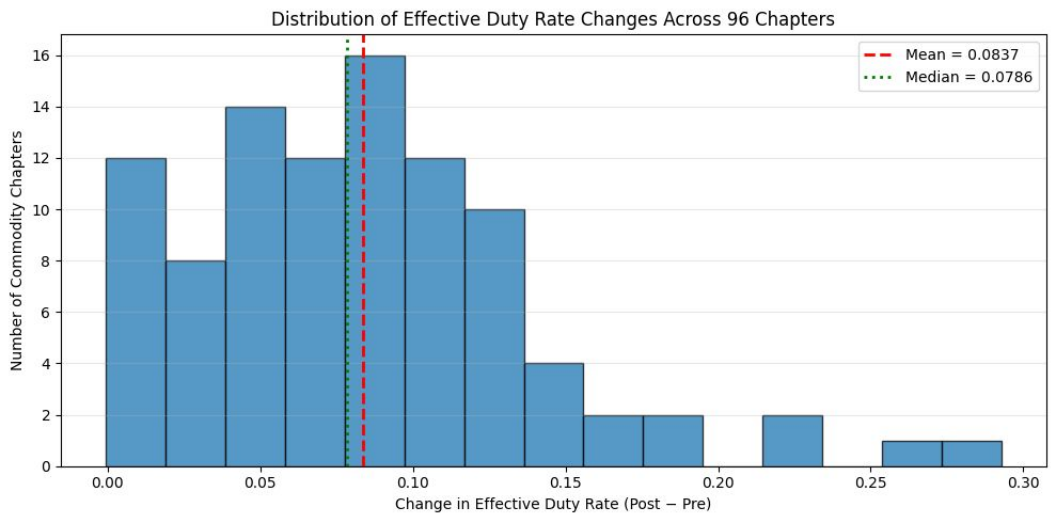


Visualisation of Change





Summary of Changes



Mean Chapter Change

+8.4pp in Effective Tariff Rate

Median Chapter Change

+7.9pp in Effective Tariff Rate

Chapters with Increase

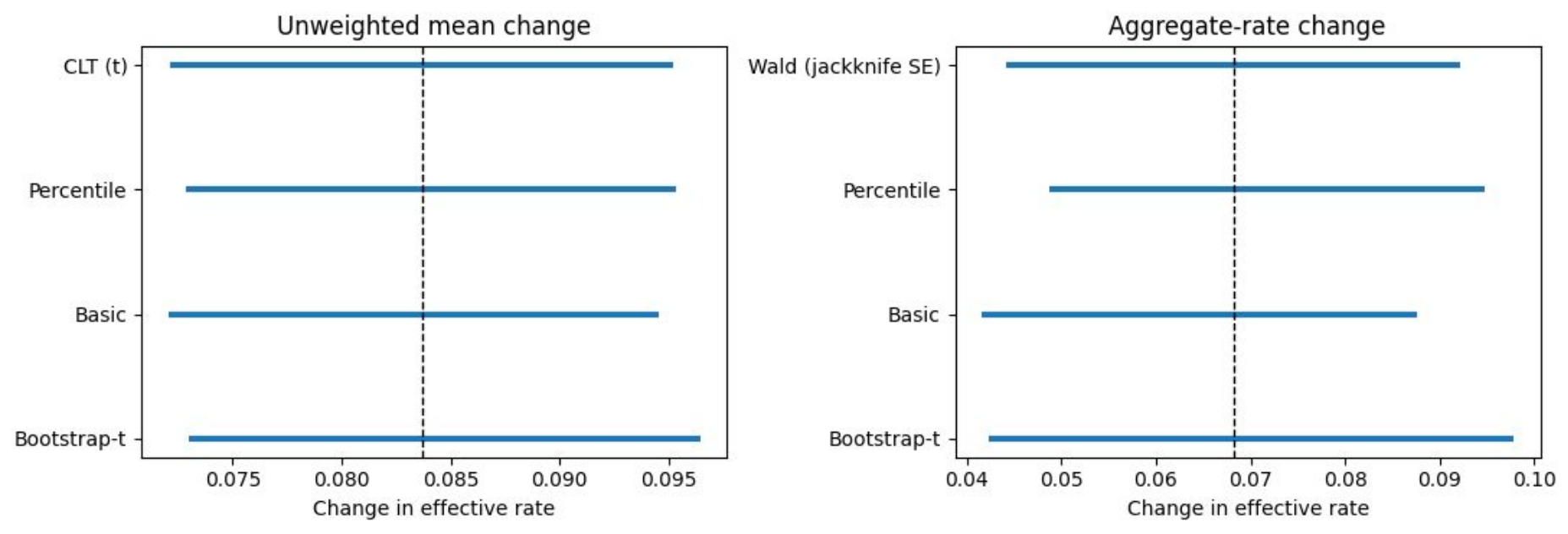
99% of Chapters faced an increase

Total No. of Chapters

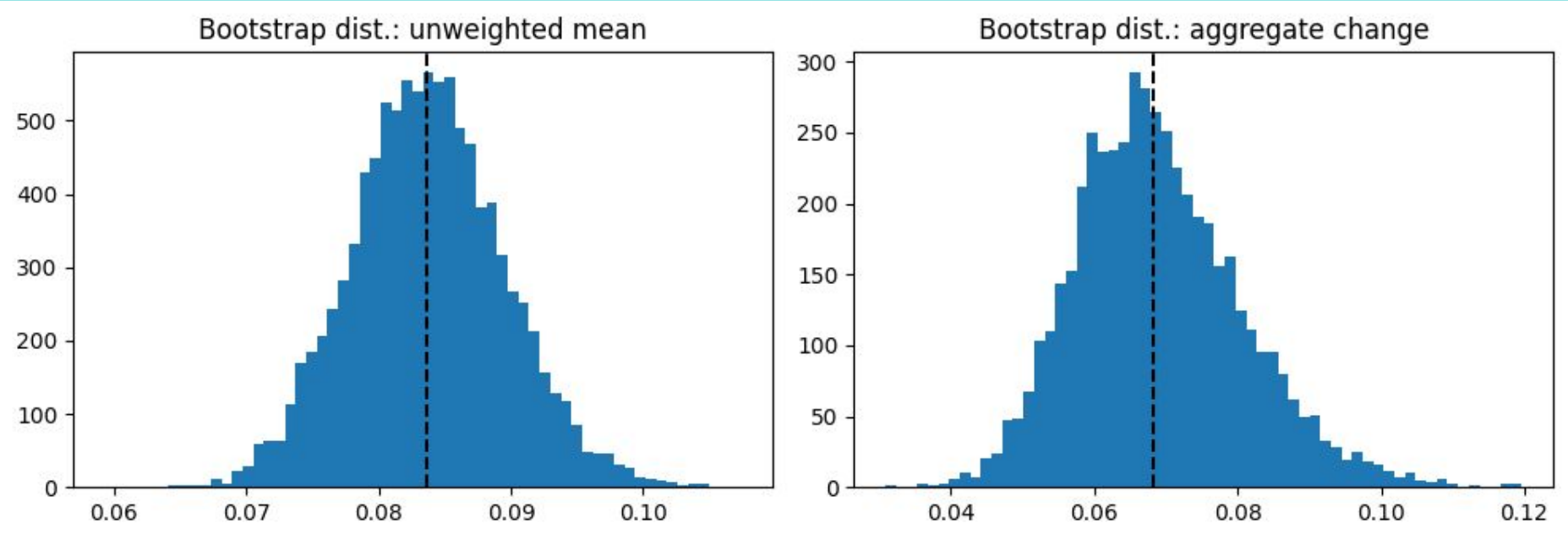
96 unique commodity chapters

1.

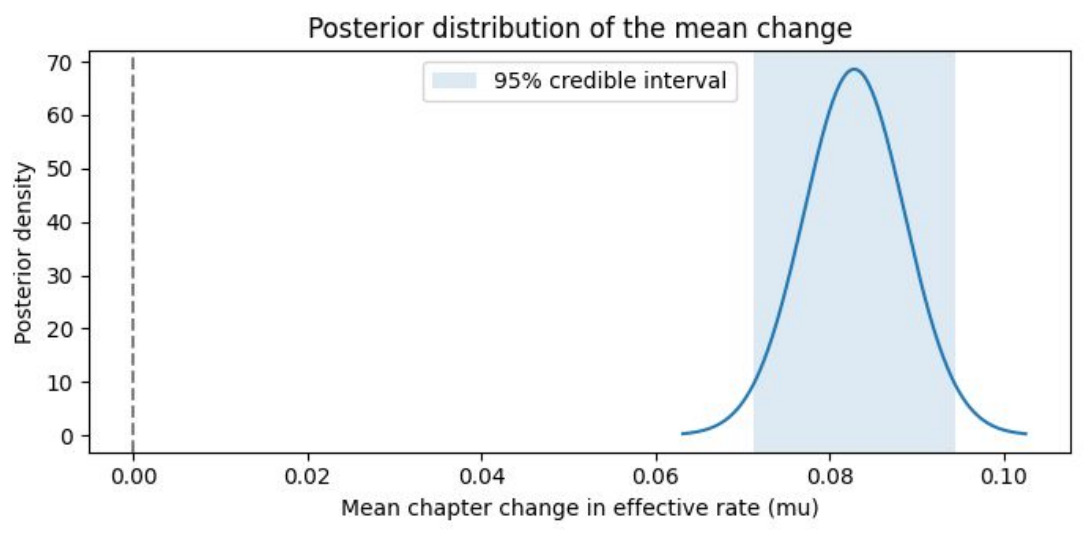
Confidence Intervals



Bootstrap Distribution Change



Bayesian Credible Interval



Prior	95% interval (pts)	P(rise > 5pts)
Flat	7.2 – 9.5	> 99.99%
Weak	7.1 – 9.4	> 99.99%
Sceptical	5.7 – 8.1	99.9%

Additional Statistical Tests

Sign-Flip Permutation Test (Mean)

Null hypothesis: The distribution of chapter-level changes is symmetric around zero.

P-value: < 0.0001

Sign Test (Median)

$H_0: P(d_i > 0) = 0.5$

P-value: 2.448624e-27

Bootstrap t-test (Mean)

$H_0: \mu_d = 0$

P-value: < 0.0001

Classic t-test (Mean)

$H_0: \mu_d = 0$

P-value: 1.301018e-25

Wilcoxon signed-rank test

Null hypothesis: The distribution of chapter-level changes is symmetric around zero.

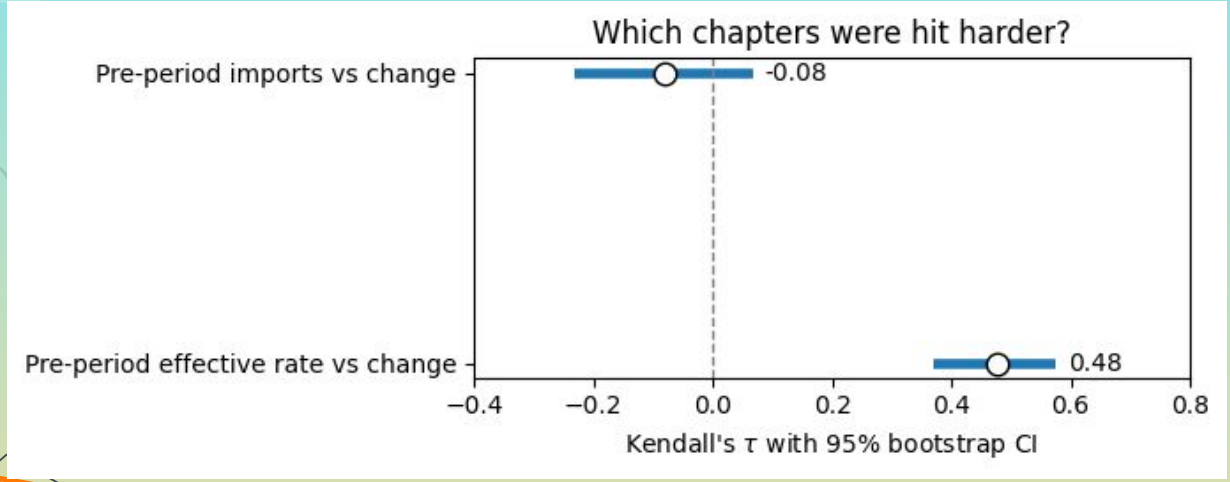
P-value: 1.838298e-17

At the 5% significance level:

- Statistically significant changes: All five tests produce p-values below 0.05.
- Average increase: The mean chapter-level change is positive at 0.0837.
- Median increase: The median chapter-level change is positive at 0.0786.
- Widespread increases: Approximately 99% of commodity chapters experienced an increase in effective duty rates.
- Robustness: The conclusions remain consistent across conventional, non-parametric and resampling-based tests.

There is strong statistical evidence that effective duty rates increased across U.S. commodity chapters between the selected pre- and post-periods.

Correlation Analysis



- Import size vs change:
  - τ = −0.08 [−0.23, 0.07].
- Starting rate vs change:
  - τ = +0.48 [0.37, 0.57]
- Kendall's τ uses rankings only, so a few huge chapters can't distort it.

Do Changes Differ Across Groups?

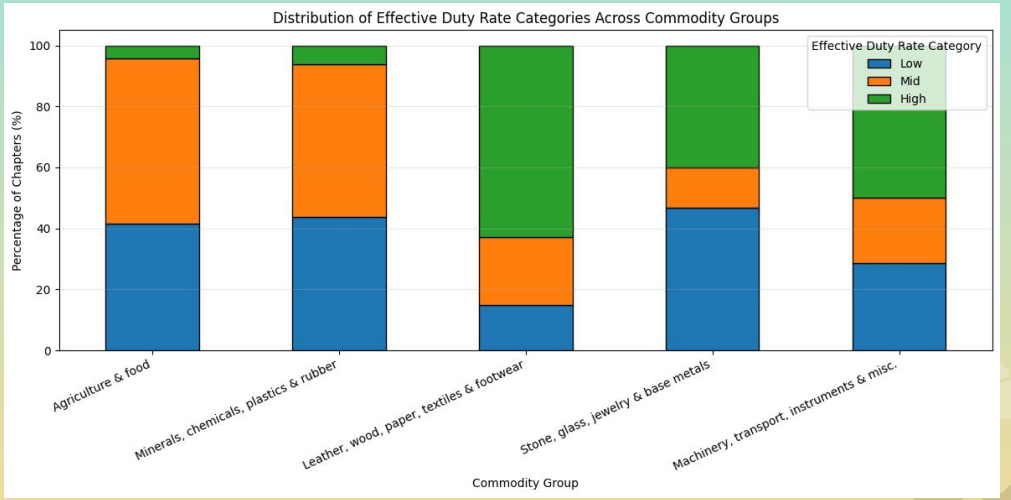
Null hypothesis: Commodity group and change category are independent

	Low	Mid	High
Agriculture & food	10	13	1
Minerals, chemicals, plastics & rubber	7	8	1
Leather, wood, paper, textiles & footwear	4	6	17
Stone, glass, jewelry & base metals	7	2	6
Machinery, transport, instruments & misc	4	3	7

Chi-Square Statistics: 30.671

P-value: 0.0002

Cramer's V: 0.400



Conclusion

Effective tariff rates increased significantly across U.S. commodity chapters.

The increase was widespread, but the magnitude differed across commodity categories.

The increase was not evenly distributed across industries, suggesting that certain commodity groups were more affected than others.





Limitations



This is a before/after comparison with no control group. It shows that rates rose, not why.



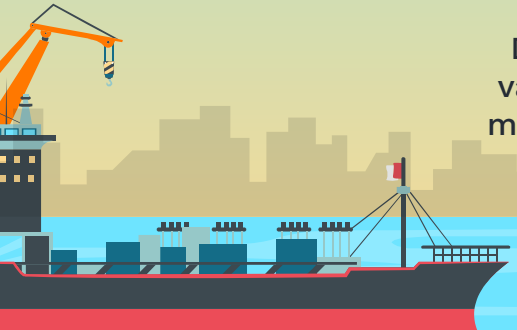
HS-2 chapters can hide large differences between products. The result covers Mar–Dec 2025 only and rates fell afterwards.



Duties collected ÷ import value also moves when the mix of products or countries changes.



The 96 chapters are the full population and are related, so very small p-values overstate certainty.

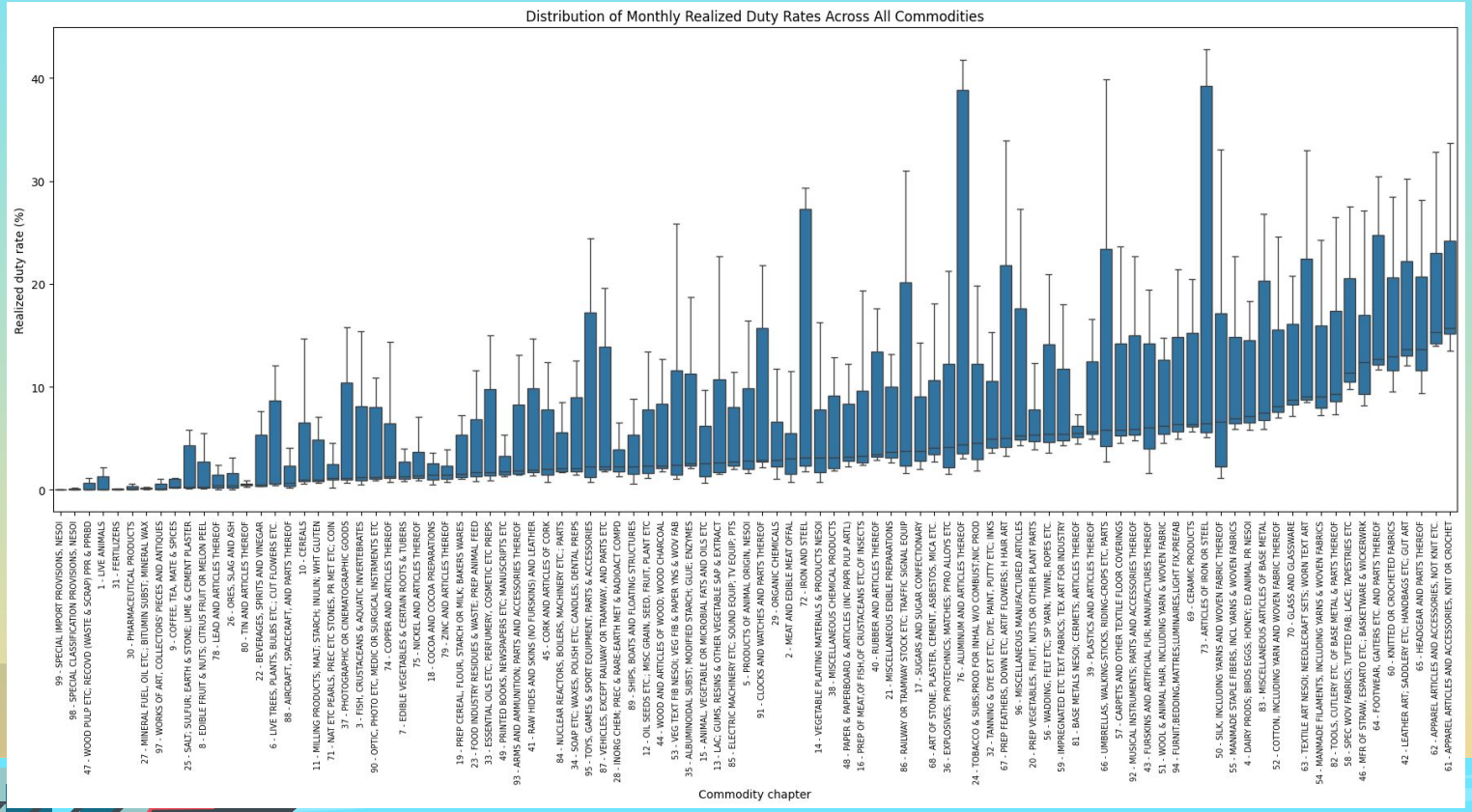


Thank You

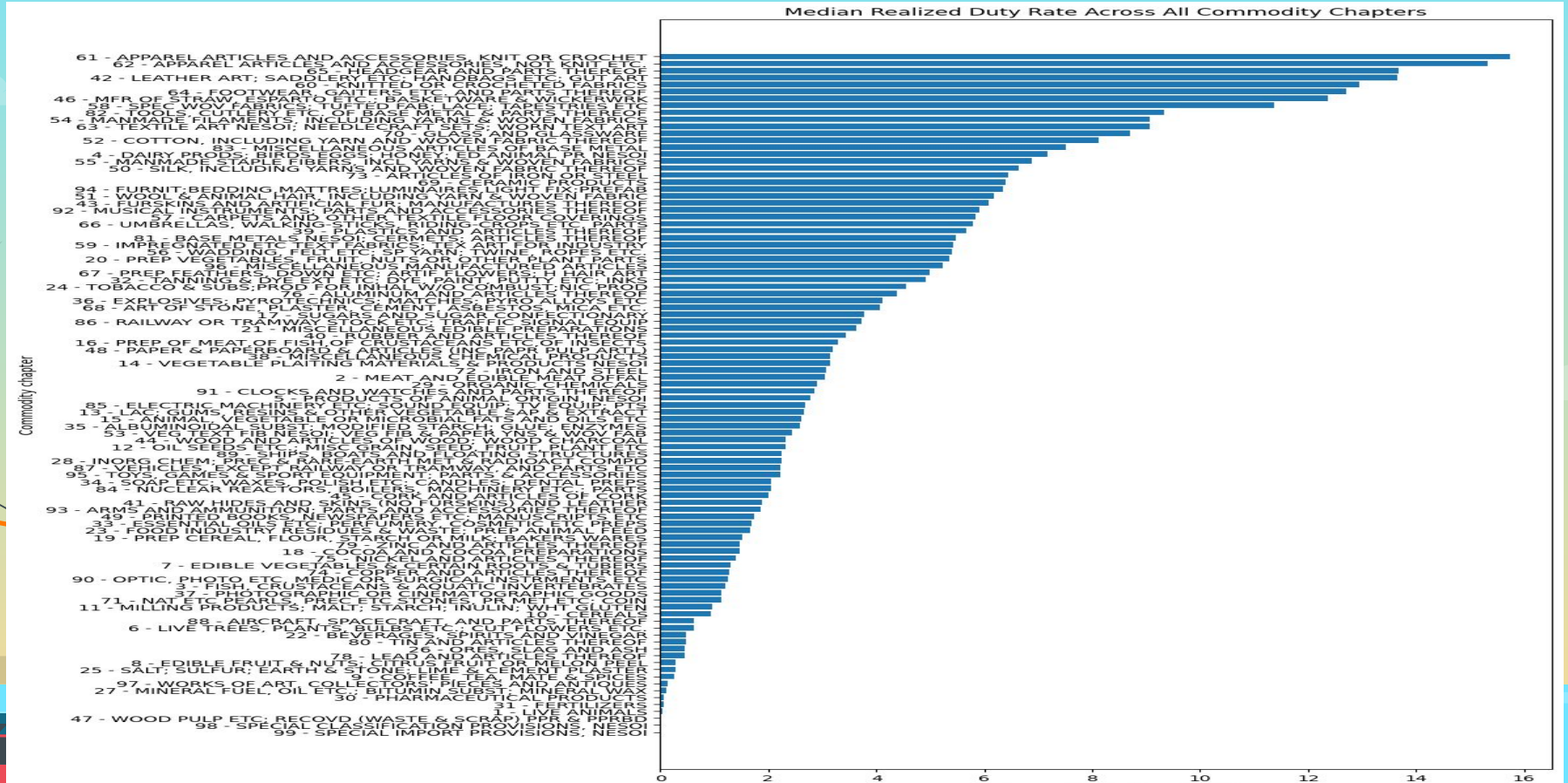




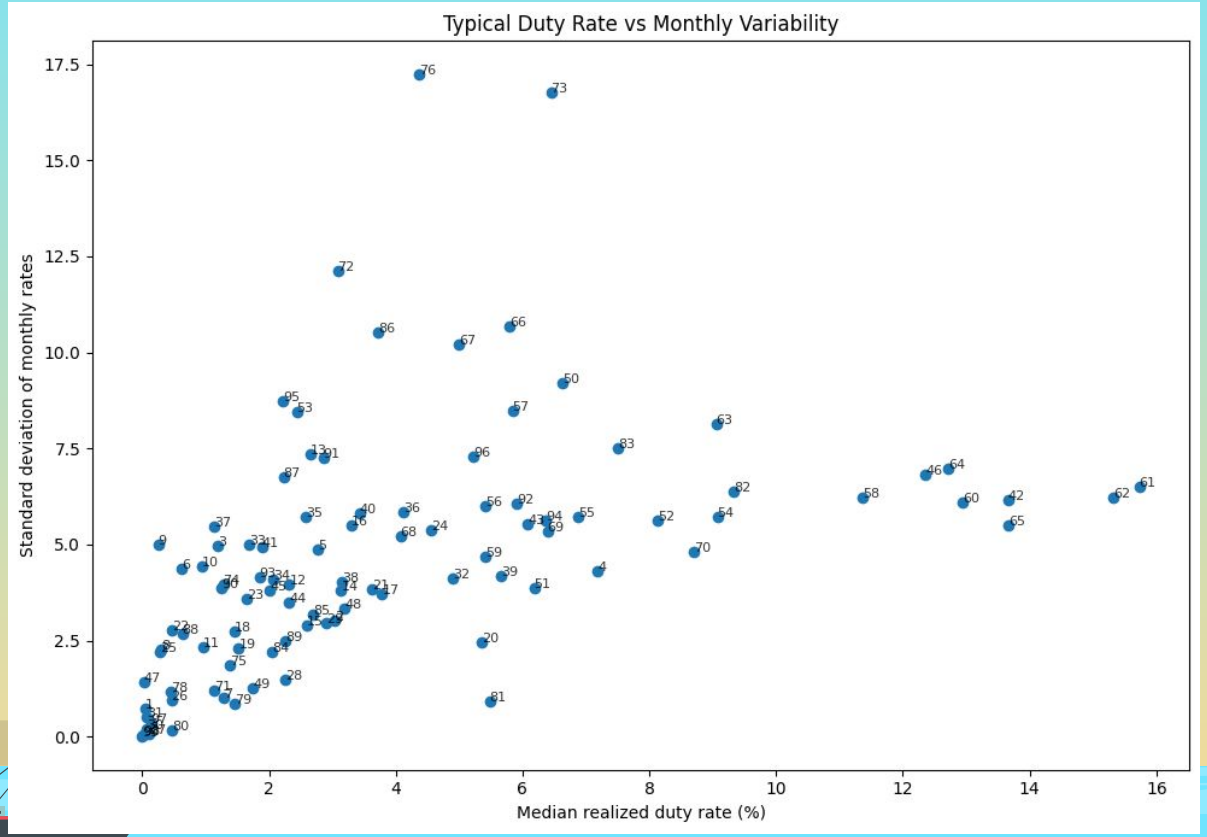
Boxplot


Distribution of Monthly Realized Duty Rates Across All Commodities
Realized duty rate (%)
0
10
20
30
40
Commodity chapter
99 - SPECIAL IMPORT PROVISIONS, NESOI
98 - SPECIAL CLASSIFICATION PROVISIONS, NESOI
47 - WOOD PULP ETC; RECOVD (WASTE & SCRAP) PPR & PPRBD
1 - LIVE ANIMALS
31 - FERTILIZERS
30 - PHARMACEUTICAL PRODUCTS
27 - MINERAL FUEL, OIL ETC.; BITUMIN SUBST; MINERAL WAX
97 - WORKS OF ART; COLLECTORS' PIECES AND ANTIQUES
9 - COFFEE, TEA, MATE & SPICES
25 - SALT; SULFUR; EARTH & STONE; LIME & CEMENT PLASTER
8 - EDIBLE FRUIT & NUTS; CITRUS FRUIT OR MELON PEEL
78 - LEAD AND ARTICLES THEREOF
26 - ORES, SLAG AND ASH
80 - TIN AND ARTICLES THEREOF
22 - BEVERAGES, SPIRITS AND VINEGAR
6 - LIVE TREES, PLANTS, BULBS ETC.; CUT FLOWERS ETC.
88 - AIRCRAFT, SPACECRAFT, AND PARTS THEREOF
10 - CEREALS
11 - MILLING PRODUCTS; MALT; STARCH; INULIN; WHT GLUTEN
71 - NAT ETC PEARLS, PREC ETC STONES, PR MET ETC; COIN
37 - PHOTOGRAPHIC OR CINEMATOGRAPHIC GOODS
3 - FISH, CRUSTACEANS & AQUATIC INVERTEBRATES
90 - OPTIC, PHOTO ETC, MEDIC OR SURGICAL INSTRMNTS ETC
74 - COPPER AND ARTICLES THEREOF
7 - EDIBLE VEGETABLES & CERTAIN ROOTS & TUBERS
75 - NICKEL AND ARTICLES THEREOF
18 - COCOA AND COCOA PREPARATIONS
79 - ZINC AND ARTICLES THEREOF
19 - PREP CEREAL, FLOUR, STARCH OR MILK; BAKERS WARES
23 - FOOD INDUSTRY RESIDUES & WASTE; PREP ANIMAL FEED
33 - ESSENTIAL OILS ETC; PERFUMERY, COSMETIC ETC PREPS
49 - PRINTED BOOKS, NEWSPAPERS; ETC; MANUSCRIPTS ETC
93 - ARMS AND AMMUNITION; PARTS AND ACCESSORIES THEREOF
41 - RAW HIDES AND SKINS (NO FURSKINS) AND LEATHER
45 - CORK AND ARTICLES OF CORK
84 - NUCLEAR REACTORS, BOILERS, MACHINERY ETC.; PARTS
34 - SOAP ETC; WAXES, POLISH ETC; CANDLES, DENTAL PREPS
95 - TOYS, GAMES & SPORT EQUIPMENT; PARTS & ACCESSORIES
87 - VEHICLES, EXCEPT RAILWAY OR TRAMWAY, AND PARTS ETC
28 - INORG CHEM; PREC & RARE-EARTH MET & RADIOACT COMPD
89 - SHIPS, BOATS AND FLOATING STRUCTURES
12 - OIL SEEDS ETC.; MISC GRAIN, SEED, FRUIT, PLANT ETC
44 - WOOD AND ARTICLES OF WOOD; WOOD CHARCOAL
53 - VEG TEXT FIB NESOI; VEG FIB & PAPER YNS & WOV FAB
35 - ALBUMINOIDAL SUBST; MODIFIED STARCH; GLUE; ENZYMES
15 - ANIMAL, VEGETABLE OR MICROBIAL FATS AND OILS ETC
13 - LAC; GUMS, RESINS & OTHER VEGETABLE SAP & EXTRACT
85 - ELECTRIC MACHINERY ETC; SOUND EQUIP; TV EQUIP; PTS
5 - PRODUCTS OF ANIMAL ORIGIN, NESOI
91 - CLOCKS AND WATCHES AND PARTS THEREOF
29 - ORGANIC CHEMICALS
2 - MEAT AND EDIBLE MEAT OFFAL
72 - IRON AND STEEL
14 - VEGETABLE PLAITING MATERIALS & PRODUCTS NESOI
38 - MISCELLANEOUS CHEMICAL PRODUCTS
48 - PAPER & PAPERBOARD & ARTICLES (INC PAPR PULP ARTL)
16 - PREP OF MEAT,OF FISH,OF CRUSTACEANS ETC,OF INSECTS
40 - RUBBER AND ARTICLES THEREOF
21 - MISCELLANEOUS EDIBLE PREPARATIONS
86 - RAILWAY OR TRAMWAY STOCK ETC; TRAFFIC SIGNAL EQUIP
17 - SUGARS AND SUGAR CONFECTIONARY
68 - ART OF STONE, PLASTER, CEMENT, ASBESTOS, MICA ETC.
36 - EXPLOSIVES; PYROTECHNICS; MATCHES; PYRO ALLOYS ETC
76 - ALUMINUM AND ARTICLES THEREOF
24 - TOBACCO & SUBS;PROD FOR INHAL W/O COMBUST;NIC PROD
32 - TANNING & DYE EXT ETC; DYE, PAINT, PUTTY ETC; INKS
67 - PREP FEATHERS, DOWN ETC; ARTIF FLOWERS, H HAIR ART
96 - MISCELLANEOUS MANUFACTURED ARTICLES
20 - PREP VEGETABLES, FRUIT, NUTS OR OTHER PLANT PARTS
56 - WADDING, FELT ETC; SP YARN; TWINE, ROPES ETC.
59 - IMPREGNATED ETC TEXT FABRICS; TEX ART FOR INDUSTRY
81 - BASE METALS NESOI; CERMETS; ARTICLES THEREOF
39 - PLASTICS AND ARTICLES THEREOF
66 - UMBRELLAS, WALKING-STICKS, RIDING-CROPS ETC.; PARTS
57 - CARPETS AND OTHER TEXTILE FLOOR COVERINGS
92 - MUSICAL INSTRUMENTS; PARTS AND ACCESSORIES THEREOF
43 - FURSKINS AND ARTIFICIAL FUR; MANUFACTURES THEREOF
51 - WOOL & ANIMAL HAIR, INCLUDING YARN & WOVEN FABRIC
94 - FURNIT;BEDDING,MATTRES;LUMINAIRES,LIGHT FIX;PREFAB
69 - CERAMIC PRODUCTS
73 - ARTICLES OF IRON OR STEEL
50 - SILK, INCLUDING YARNS AND WOVEN FABRIC THEREOF
55 - MANMADE STAPLE FIBERS, INCL YARNS & WOVEN FABRICS
4 - DAIRY PRODS; BIRDS EGGS; HONEY; ED ANIMAL PR NESOI
83 - MISCELLANEOUS ARTICLES OF BASE METAL
52 - COTTON, INCLUDING YARN AND WOVEN FABRIC THEREOF
70 - GLASS AND GLASSWARE
63 - TEXTILE ART NESOI; NEEDLECRAFT SETS; WORN TEXT ART
54 - MANMADE FILAMENTS, INCLUDING YARNS & WOVEN FABRICS
82 - TOOLS, CUTLERY ETC. OF BASE METAL & PARTS THEREOF
58 - SPEC WOV FABRICS; TUFTED FAB; LACE; TAPESTRIES ETC
46 - MFR OF STRAW, ESPARTO ETC.; BASKETWARE & WICKERWRK
64 - FOOTWEAR, GAITERS ETC. AND PARTS THEREOF
60 - KNITTED OR CROCHETED FABRICS
42 - LEATHER ART; SADDLERY ETC.; HANDBAGS ETC; GUT ART
65 - HEADGEAR AND PARTS THEREOF
62 - APPAREL ARTICLES AND ACCESSORIES, NOT KNIT ETC.
61 - APPAREL ARTICLES AND ACCESSORIES, KNIT OR CROCHET

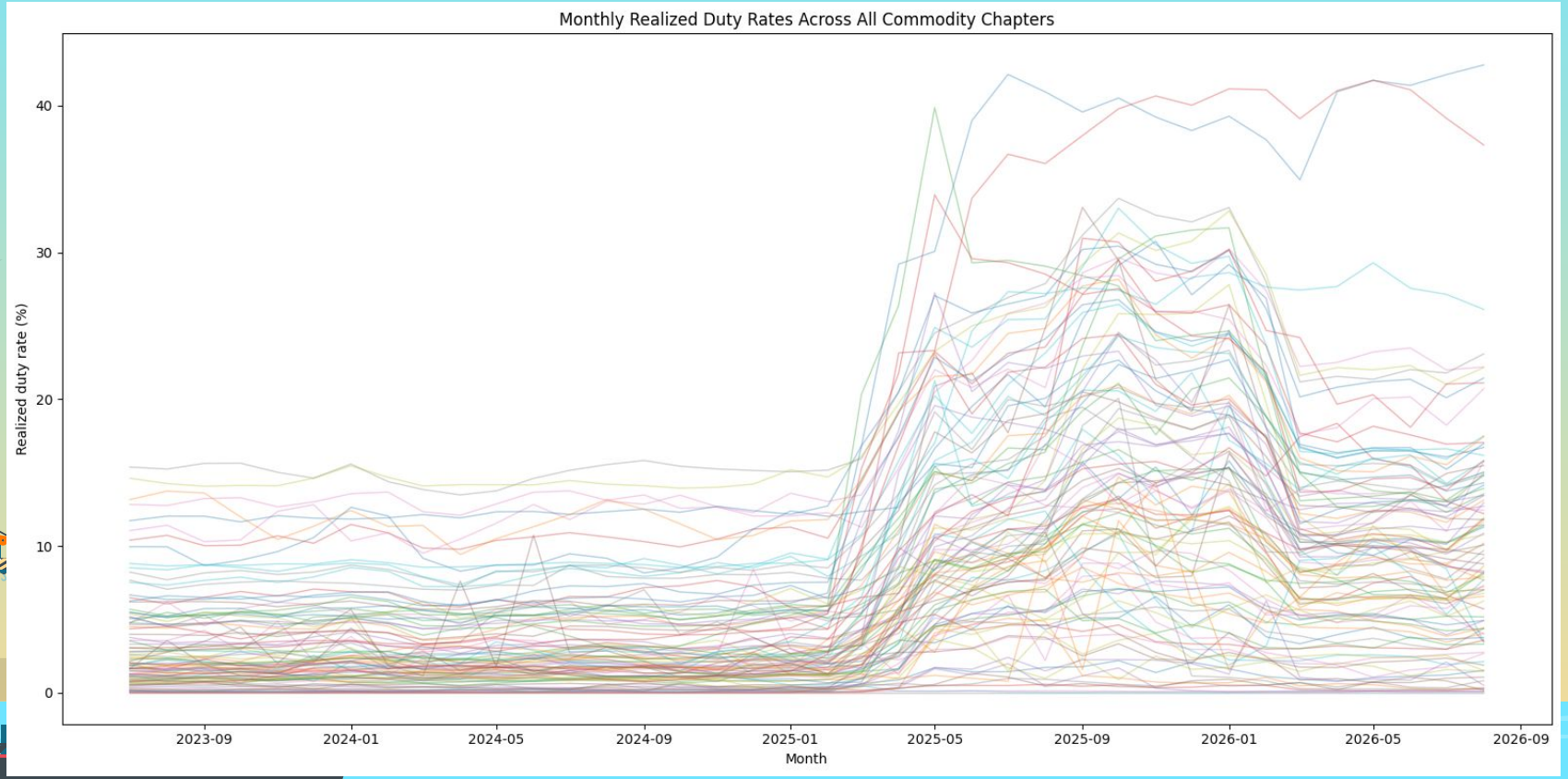
Bar Graph


Median Realized Duty Rate Across All Commodity Chapters
Commodity chapter
61 - APPAREL ARTICLES AND ACCESSORIES, KNIT OR CROCHET
62 - APPAREL ARTICLES AND ACCESSORIES, NOT KNIT ETC
65 - HEADGEAR AND PARTS THEREOF
42 - LEATHER ART; SADDLERY ETC; HANDBAGS ETC; GUT ART
60 - KNITTED OR CROCHETED FABRICS
64 - FOOTWEAR, GAITERS ETC. AND PARTS THEREOF
46 - MFR OF STRAW, ESPARTO ETC; BASKETWARE & WICKERWRK
58 - SPEC WOV FABRICS; TUFTED FAB; LACE; TAPESTRIES ETC
82 - TOOLS, CUTLERY ETC. OF BASE METAL & PARTS THEREOF
54 - MANMADE FILAMENTS, INCLUDING YARNS & WOVEN FABRICS
63 - TEXTILE ART NESOI; NEEDLECRAFT SETS; WORN TEXT ART
70 - GLASS AND GLASSWARE
52 - COTTON, INCLUDING YARN AND WOVEN FABRIC THEREOF
83 - MISCELLANEOUS ARTICLES OF BASE METAL
4 - DAIRY PRODS; BIRDS EGGS; HONEY; EDIBLE ANIMAL PR NESOI
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39 - PLASTICS AND ARTICLES THEREOF
81 - BASE METALS NESOI; CERMETS; ARTICLES THEREOF
59 - IMPREGNATED, ETC TEXT FABRICS; TEX ART FOR INDUSTRY
56 - WADDING, FELT ETC; SP YARN; TWINE, ROPES, ETC.
20 - PREP VEGETABLES, FRUIT, NUTS OR OTHER PLANT PARTS
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24 - TOBACCO & SUBS;PROD FOR INHAL W/O COMBUST;NIC PROD
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36 - EXPLOSIVES; PYROTECHNICS; MATCHES; PYRO ALLOYS; ETC
68 - ART OF STONE, PLASTER, CEMENT, ASBESTOS, MICA ETC.
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98 - SPECIAL CLASSIFICATION PROVISIONS, NESOI
99 - SPECIAL IMPORT PROVISIONS, NESOI
0
2
4
6
8
10
12
14
16

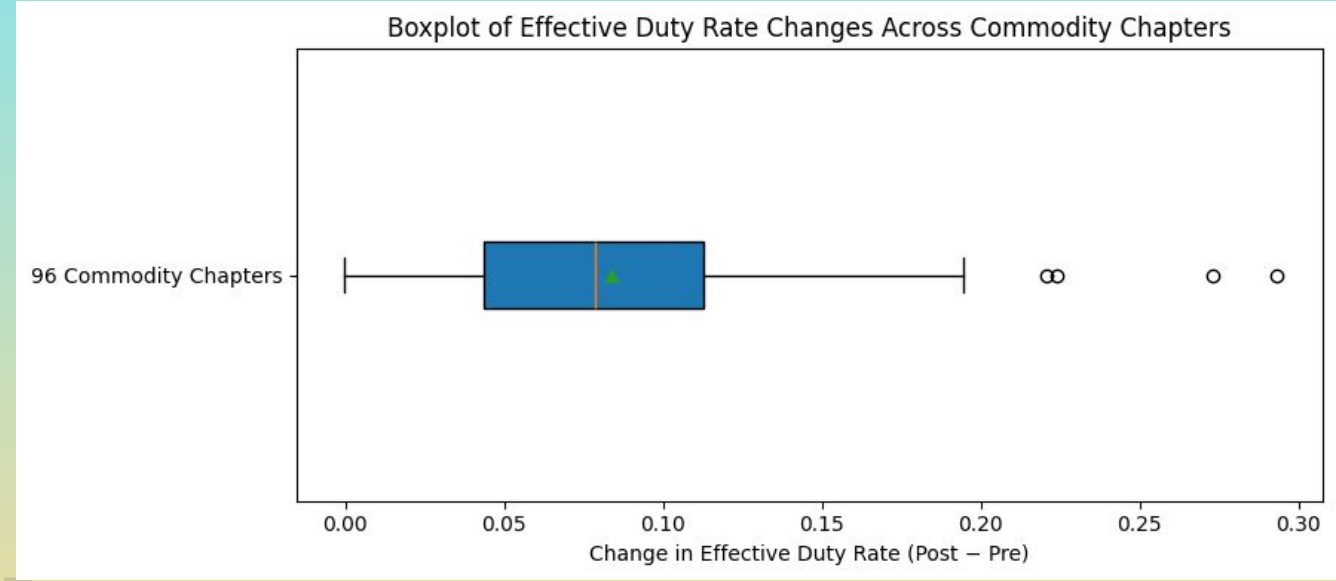
Scatter Plot



Line Graph


Monthly Realized Duty Rates Across All Commodity Chapters
Realized duty rate (%)
0
10
20
30
40
2023-09
2024-01
2024-05
2024-09
2025-01
2025-05
2025-09
2026-01
2026-05
2026-09
Month

Boxplot


Boxplot of Effective Duty Rate Changes Across Commodity Chapters
96 Commodity Chapters
0.00
0.05
0.10
0.15
0.20
0.25
0.30
Change in Effective Duty Rate (Post − Pre)

Scatter Plot

